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PyIU Python Crash Course

AN INTRODUCTION TO PYTHON FOR STUDENTS INTERESTED IN ASTRONOMY/STEM

By: IU Astronomy & Physics Graduate Students

10/12/26

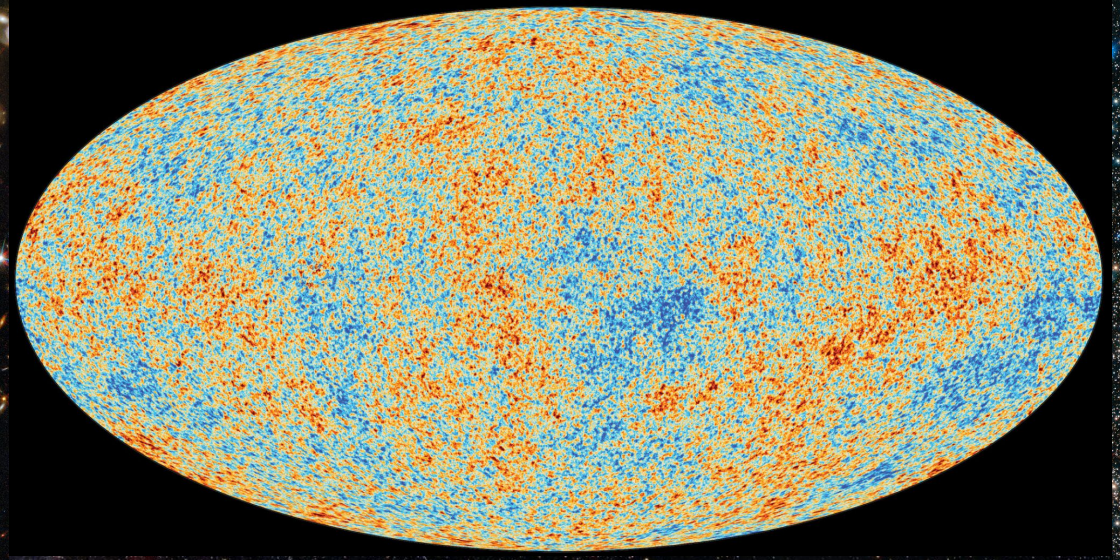
Crash Course Purpose

- Astronomers, physicists, and other scientists use **programming** as a tool to analyze data for research
- Python has become the most-used programming language, becoming a vital skill!



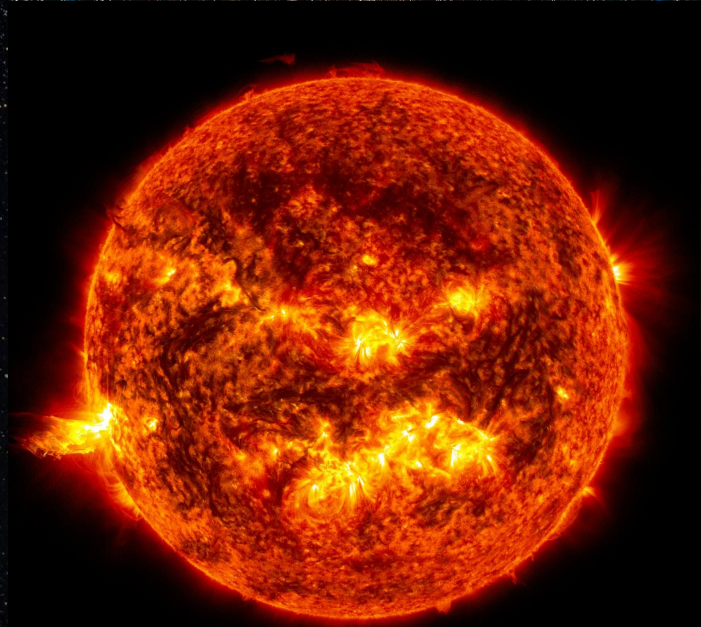
But Python is much more user-friendly than these!
(at the cost of being a bit slower)

Other common languages:
C, C++, C#
Java & JavaScript
SQL
R
MATLAB
Rust
Perl
HTML & CSS
Fortran
Julia

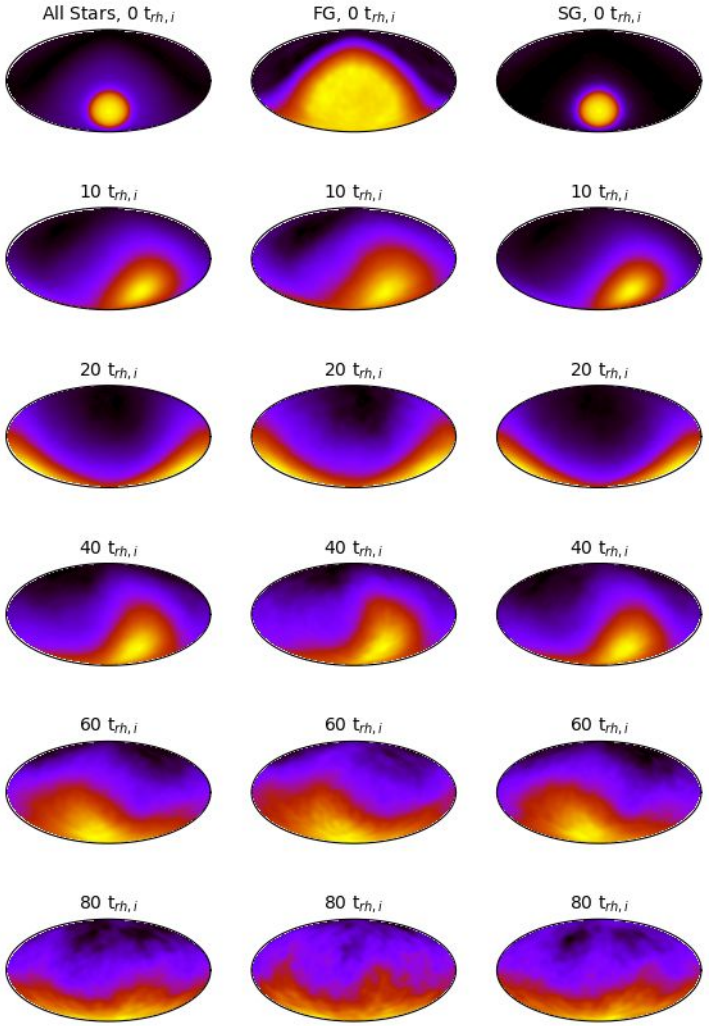



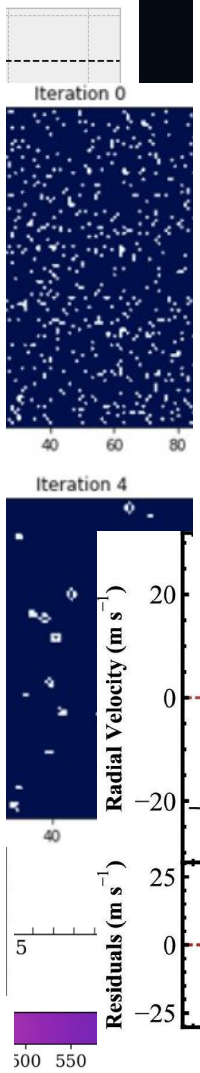

Coding is your gateway to the universe...

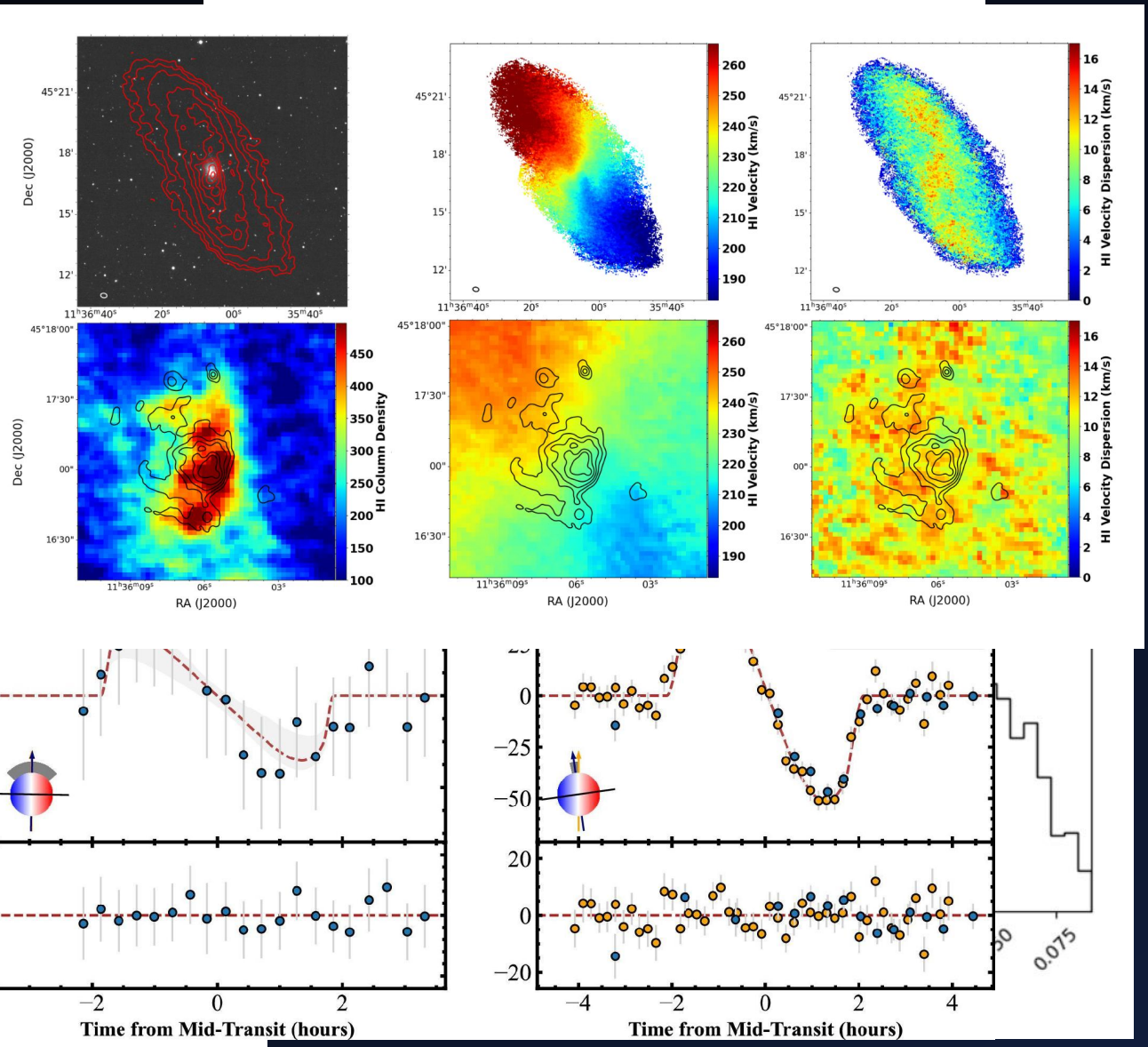




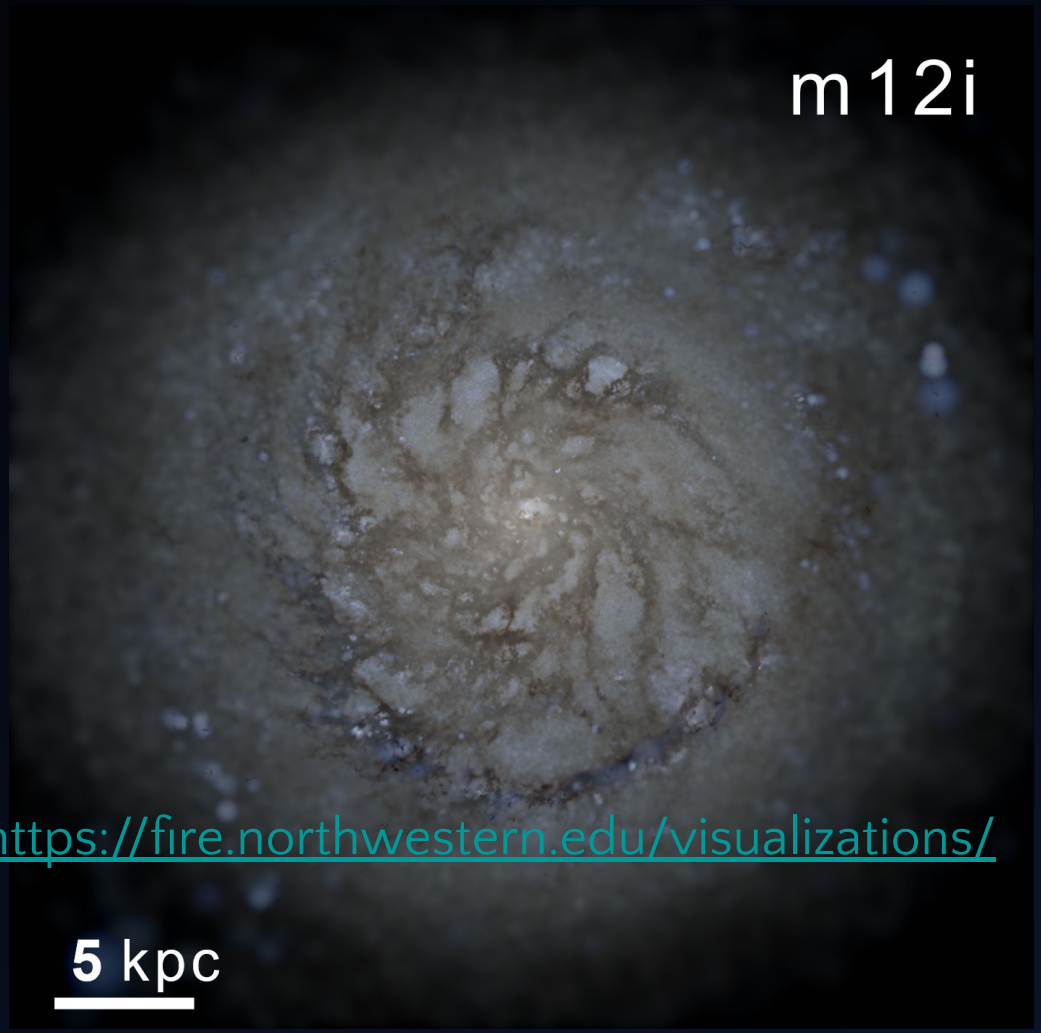
What we've done with Python!



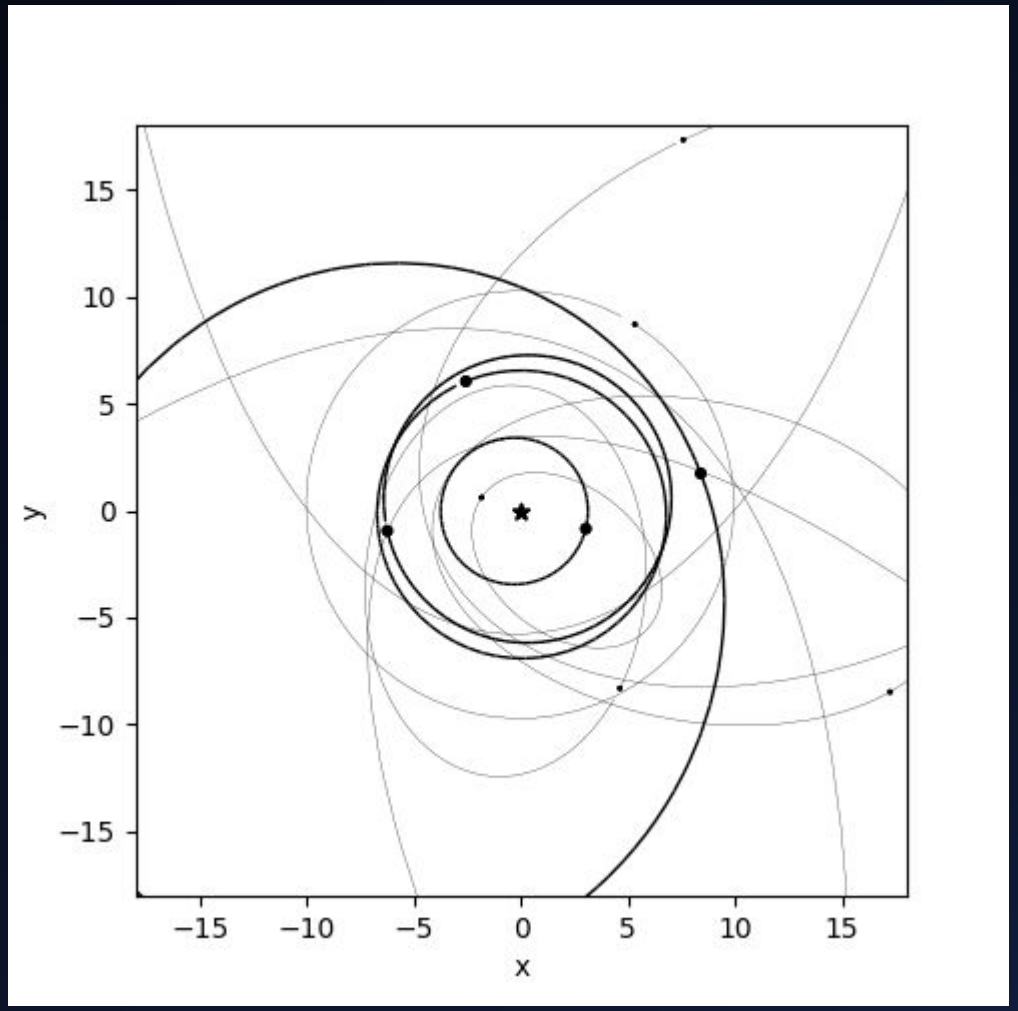




More Examples of Python from IU Astronomy



https://fire.northwestern.edu/visualizations/



Python



Pros of Python:

- Simple and **user-friendly**
- Interpreted, high-level language

Cons:

- A bit of a black box, **slow**
- Interpreted, high-level language

Documentation:
https://docs.python.org/3/

Lots of “layers” to executing the code, computes line-by-line

Easily readable to humans… not so much for a computer

A Demonstrative Example: “Hello World”





Where to write code in Python



Where to write code in Python



Where to write code in Python

- IDEs: Integrated Development Environments





PyCharm

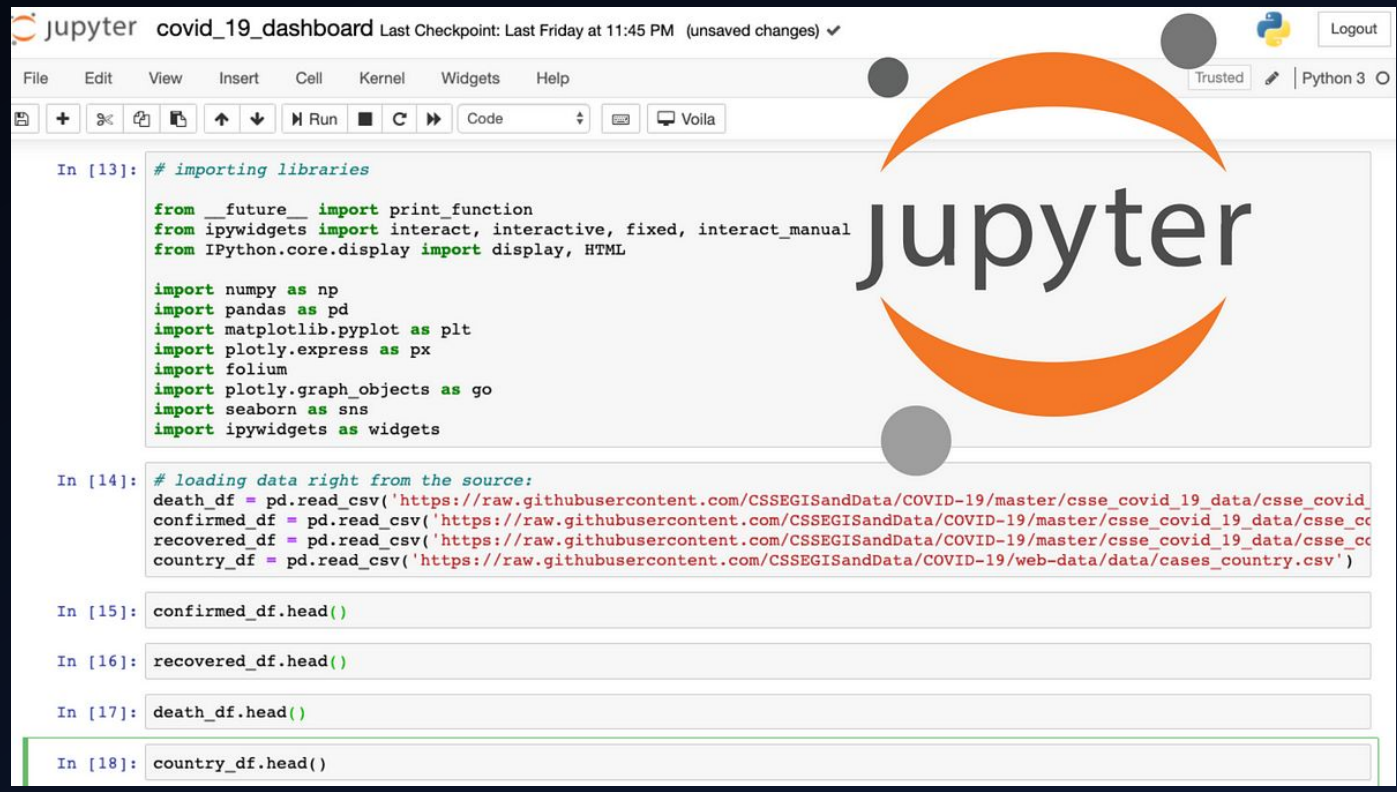




VSCode

What we're using for the Crash Course

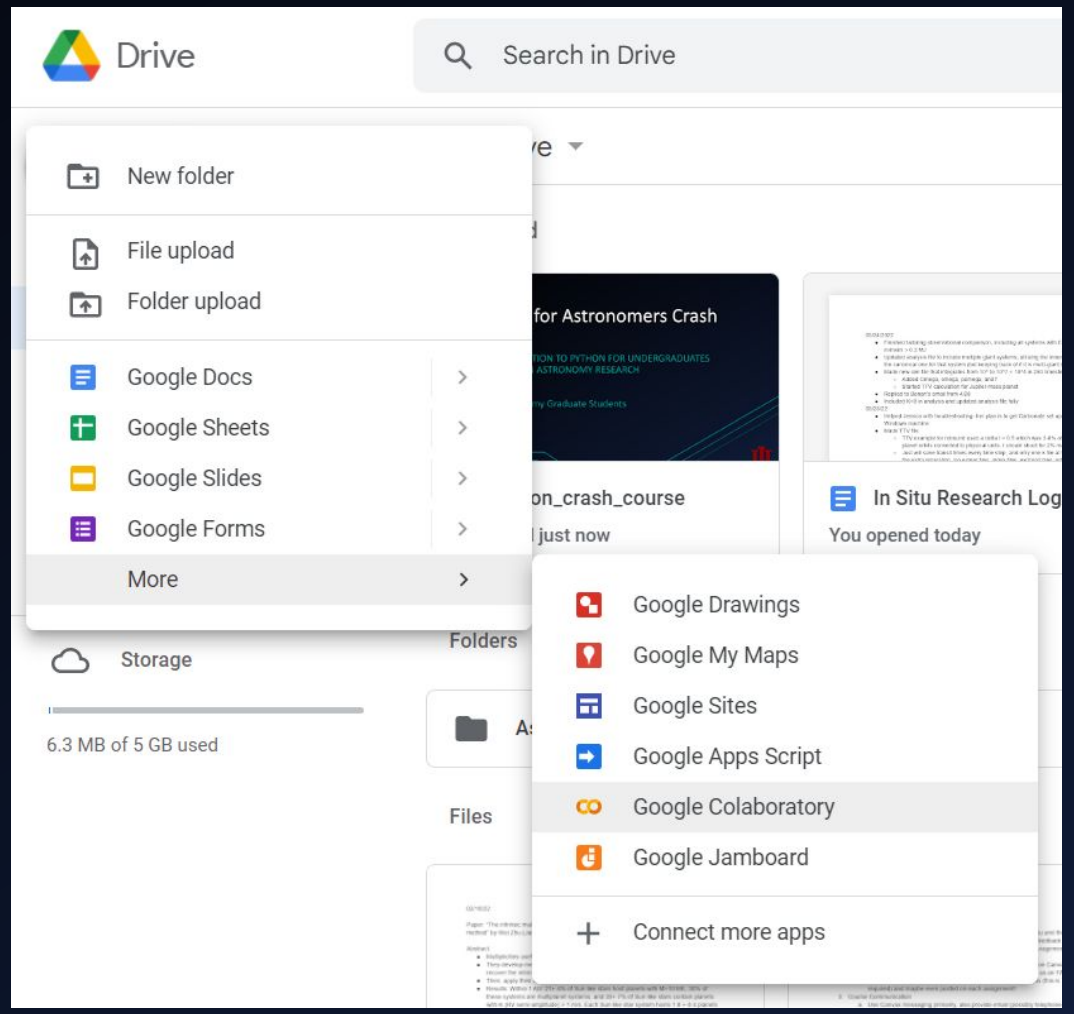
- **Jupyter notebook**: web-based interactive Python development tool
- **Cell-by-cell code blocks** allow for:
  - Easy debugging & visualization
  - Quick and efficient code development



How we're accessing Jupyter: Google "Colab"oratory

colab.research.google.com

- Free web-based access to Jupyter notebooks
- Contains searchable practice examples and datasets
- Allows collaboration* **(Google docs but for coding)**
  - *asynchronously



The H-R Diagram and NGC 2420

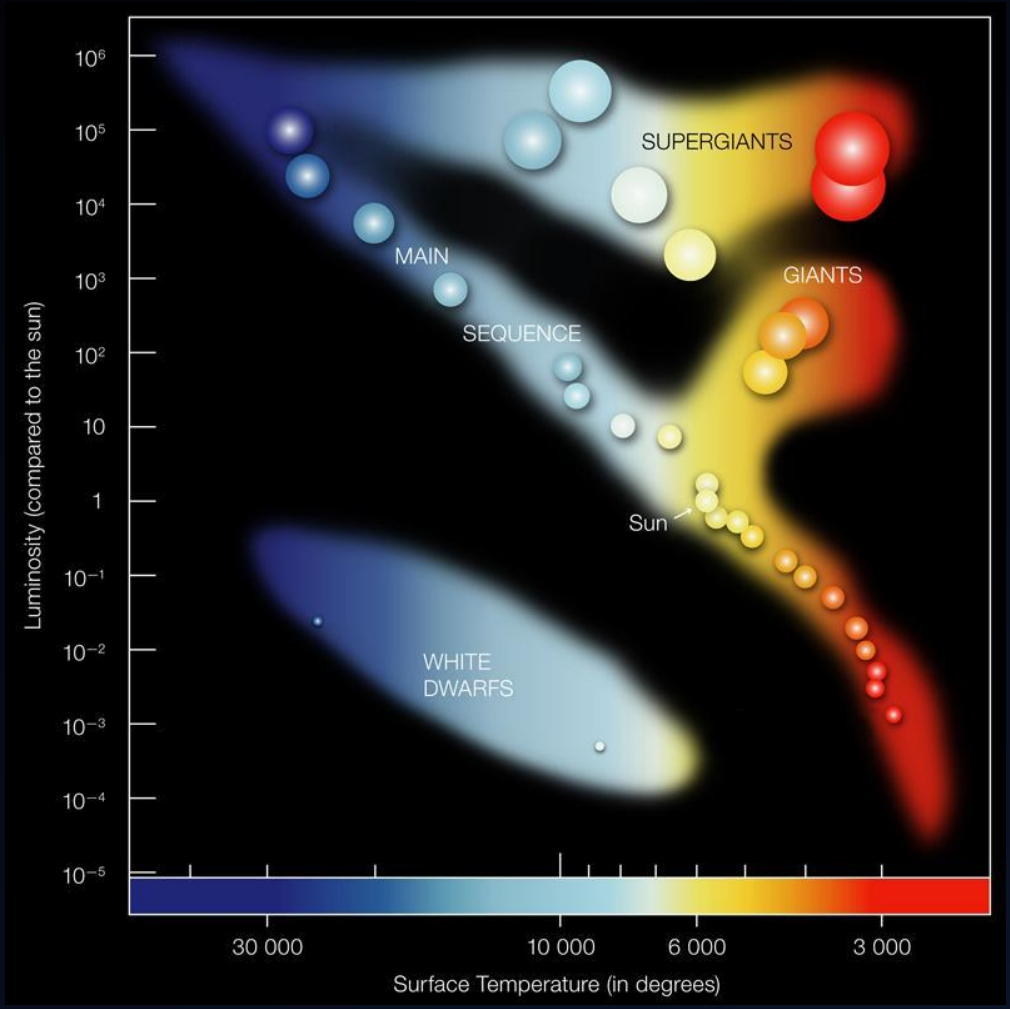
PyIU Python Crash Course

Sarah Popp



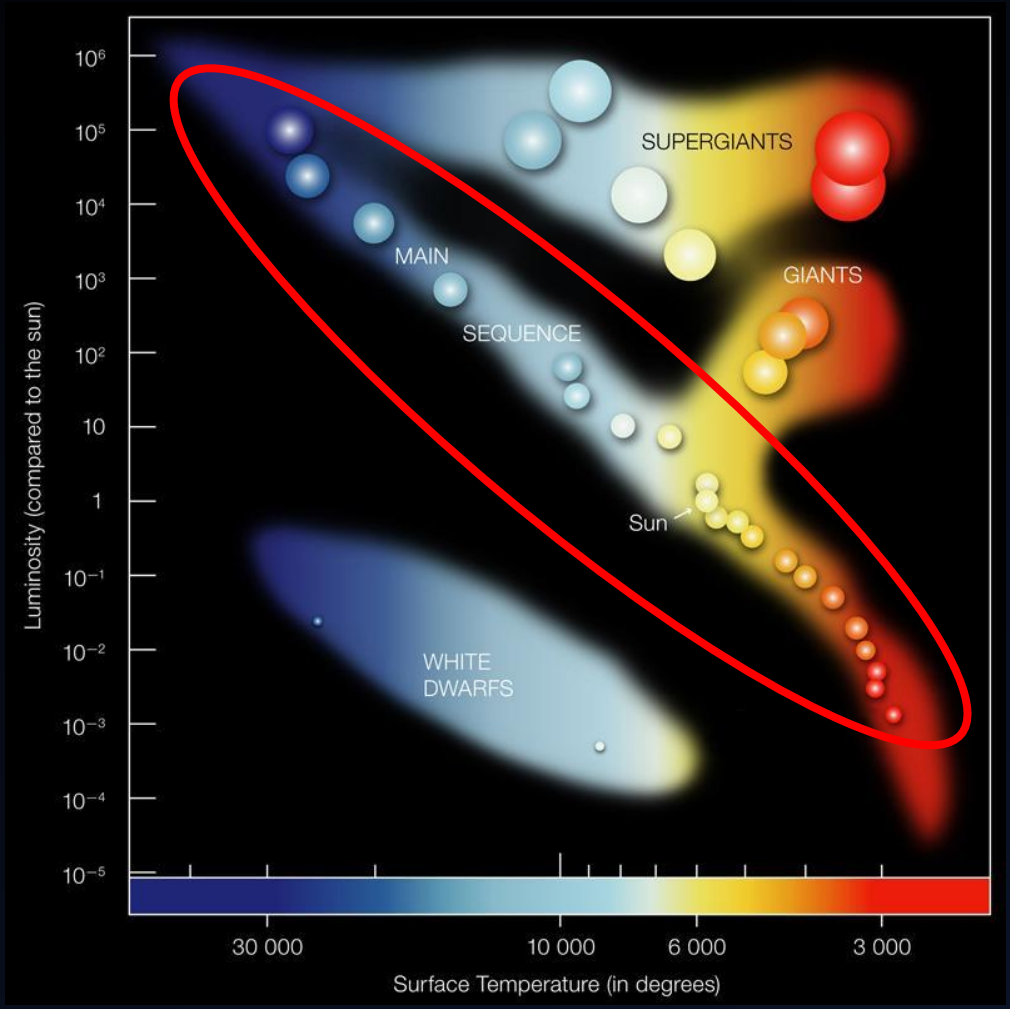


The H-R Diagram



- 1911: Ejnar Hertzsprung
- 1913: Henry Norris Russell
- **Temperature and luminosity of stars are related**
- Indicative of a star's age and evolutionary state

The H-R Diagram



- Stars spend most of their life on the **main sequence**
- All stars follow the same basic evolutionary sequence over different timescales:
  - Pre-main sequence
  - Main sequence
  - Giant
  - Death

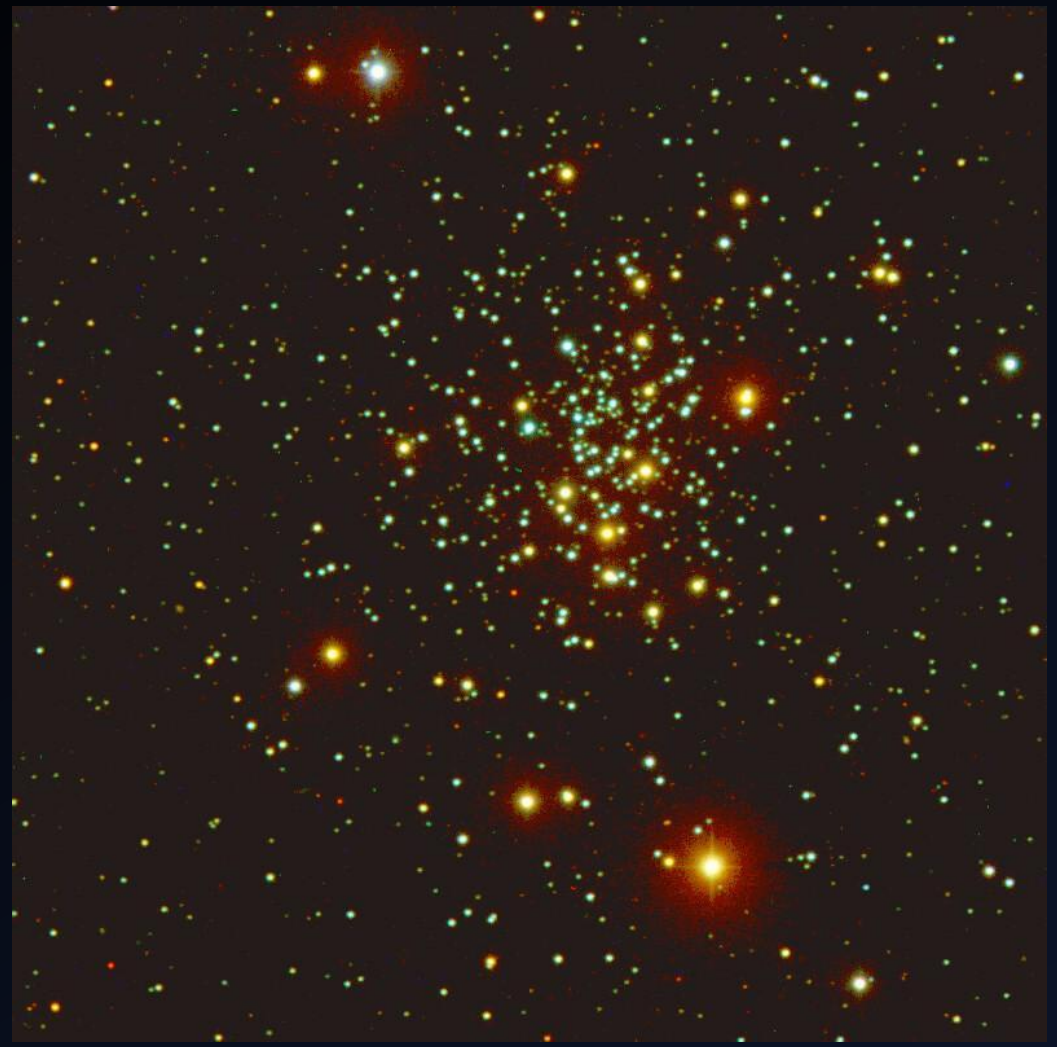
Magnitudes and Colors

- Inverse logarithmic scale
  - Additive difference in magnitudes = multiplicative difference in brightness
  - A star of magnitude 12 is 2.5 times brighter than a star of magnitude 13
- **Smaller magnitudes** correspond to **brighter stars**
- **Larger magnitudes** correspond to **fainter stars**
- Combine **apparent magnitude (*m*)** and **distance in parsecs (*d*)** to calculate **absolute magnitude (*M*)**:
  - $m - M = 5 \log (d/10)$

Magnitudes and Colors

- **Color** is the difference between magnitudes in two different filters
  - "Filter" = a region of the electromagnetic spectrum (e.g. blue light, green light, infrared light)
- Example: $B - V$ color
- **More negative colors** indicate a **hotter, bluer star**
- **More positive colors** indicate a **cooler, redder star**

NGC 2420



- Real data taken by IU astronomy grad students
- Old, metal-poor cluster
  - Metals = elements other than hydrogen and helium
  - Metal abundances less than the Sun
- Apparent V magnitudes and B-V colors
- Learn Python and construct your own HR diagram!

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Let’s Code!



Advanced Python Workshop: Coming November 11th!

- Learn techniques that you can apply directly to your **research/upper-level coursework**
- Casual, highly interactive environment with experienced instructors
- Potential topics:
  - Python installation/environments
  - Machine learning
  - Advanced plotting
  - Statistics
  - Numerical integration & simulations
  - Big data analysis & data fitting
  - **YOU tell US!**

